

Capital Inflows and Fund Economics

Research report · 7 October 2026 USD 7 million of initial owner capital · Historical and hypothetical fund experiments

Abstract

This study examines how investor subscriptions, redemptions, fees and outside-fund costs affect fund assets, owner invested equity and manager cash. The historical branch executes the MSCI World factor portfolio with absolute ± 5 -percentage-point sleeve bands and separate 1.25x leverage management over 3 October 2014-3 October 2024. Its steady-acquisition unit time-weighted return is 10.7339% annually, versus 11.0797% for a diagnostic that omits portfolio flow feedback. Separate flat, fixed-growth and imposed-loss scenarios test adjustable business assumptions. Fundraising increases externally owned assets without automatically improving owner wealth or funding the manager. At the assumed 0.05% (5 basis points) investor charge, all retained historical manager labels have negative terminal external-business cash under the stated budgets. These experiments are conditional calculations rather than fundraising or return forecasts. [Fund return comparison: historical_inflow_interpretation.compare_accounts](#) · [Client, fee and cost settings: inflow_acquisition.json](#).

1. Research Question and Economic Framework

How much externally owned capital can specified client-acquisition schedules generate, and would external fee receipts cover the assumed business costs? How do returns, redemptions and subscription timing change that answer?

Four quantities must remain separate. **Owner invested equity** is the owner's fund holding. **External invested equity** belongs to subscribing clients. **Total net fund assets** combine the two invested cohorts. **Manager cash** is held outside the fund and reflects fee receipts less the stated business costs. Assets raised from clients do not become owner wealth, and investment performance does not establish manager profitability.

Subscriptions buy units; redemptions return cash to investors. Closing external equity excludes cash already redeemed. Ending fund assets alone therefore cannot rank investor outcomes. Owner fees transferred to an owner-associated manager are internal transfers within the assumed owner group; they are not new externally earned revenue.

2. Assumptions and Accounting

AI-assisted methodology and writing: OpenAI Codex (OpenAI, n.d.) assisted development of the historical and hypothetical model code and all narrative sections, tables and figures. The final AI Assistance and Responsibility section identifies the affected material, tools, author decisions, checks and retained prompt evidence.

2.1 Client acquisition and retention

The central ticket is USD 250,000. Client equivalents may be fractional for monthly modelling and may include repeat subscriptions. They are proposed inputs, not signed clients or evidence of a sales funnel. First-year subscriptions begin in month 7 and are distributed over months 7-12. [Client, fee and cost settings: inflow_acquisition.json](#).

Table 1. Acquisition schedules before loss-year cuts · [Client schedules: inflow_acquisition.json](#)

Case	Annual new client equivalents, years 1-10	Scheduled ten-year gross subscriptions (USD m), before stress cuts
No external flows	0, 0, 0, 0, 0, 0, 0, 0, 0, 0	0.00
Limited demand	0, 1, 1, 2, 2, 2, 2, 2, 2, 2	4.00
Steady acquisition	2, 4, 6, 8, 10, 12, 14, 16, 18, 20	27.50
Strong acquisition	4, 8, 12, 16, 20, 24, 28, 32, 36, 40	55.00
Sales stop / runoff	4, 4, 0, 0, 0, 0, 0, 0, 0, 0	2.00

Ordinary cases use 10% annual external-unit attrition; sales-stop/runoff uses 20%. Each annual fraction becomes a constant equivalent monthly fraction. This is a fraction of existing external units, not the same fraction of annual subscriptions. Newly subscribed units can redeem from the following month. [Client, fee and cost settings: inflow_acquisition.json](#).

The schedules, ticket sizes and retention assumptions are not empirically calibrated. The strong schedule requires 40 new client equivalents in the final year. Contacts, conversion rates, servicing capacity, client tickets and redemption evidence are needed before adopting a commercial plan. [Client, fee and cost settings: inflow_acquisition.json](#).

2.2 Fees and costs

Table 2. Central business assumptions and sensitivities · [Business costs and fees: inflow_acquisition.json](#)

Input	Central discussion assumption
Annual fixed manager cost	USD 25,000 in year 1, rising 2% annually
Setup cost	USD 25,000 paid in month 1
Acquisition spending	0.50% of gross subscriptions
Servicing cost	0.02% annually of opening external assets
Manager receives investor fee	100% centrally; 50% sensitivity
Investor fee	0.05% centrally; 0.10%, 0.25%, 0.50% sensitivities

The investor fee is a charge in the simulation, not a verified total expense ratio or necessarily the amount retained by the manager. Embedded investment-product costs, portfolio trading and financing, the added fund fee and outside-fund business costs are separate layers.

Business costs are paid outside the fund. They reduce manager cash rather than unit NAV. If any of these costs are instead charged to the fund, investor accounts require recalculation. The budget is illustrative, may overlap bundled provider invoices and includes no full-time salary provision. Taxes, unexpected legal costs and borrowing interest on manager deficits are excluded. A funding gap is an additional working-capital requirement, not an automatic withdrawal from the owner's invested capital.

2.3 Two distinct return experiments

Historical branch: actual market observations drive the same 60/15/10/15 portfolio, absolute ± 5 -percentage-point bands, separate 1.25x leverage with ± 0.10 adjustments, and investment costs and financing used by the companion backtest. Its horizon is **3 October 2014-3 October 2024**, comprising 120 project months. Fees, client schedules, redemptions and budgets are hypothetical. Acquisition plans and attrition are fixed; they do not adopt the annual-loss demand rule below. [Fund return comparison: historical_inflow_interpretation.compare_accounts](#).

Hypothetical controls: 0% throughout; 6% throughout; -30% in year 1 then 0%; and 6% in years 1-3, -30% in year 4 then 0%. These are nominal returns after underlying investment costs but before the added fund fee, with no actual leveraged-portfolio replay. Annual returns become constant monthly returns through $(1 + \text{annual return})^{(1/12)} - 1$; the monthly fee is deducted separately from opening assets. Losses are therefore smoothed within their year unless the loss-placement control is used. [Client, fee and cost settings: inflow_acquisition.json](#).

In hypothetical acquisition loss years, subscriptions are halved and annual external-unit attrition rises to 30%. This is an imposed concurrent market-and-demand stress rather than a predictive rule. The separate timing experiment holds gross schedules and 10% annually attrition fixed even in loss years, so it answers a different question. None of these return paths or demand rules is a forecast or worst-case bound. [Client, fee and cost settings: inflow_acquisition.json](#) · [Subscription schedules: inflow_timing.json](#).

2.4 Accounting sequence

In the hypothetical controls, each month applies the investment return and fee to opening assets, then redeems existing external units and issues new units. End-month subscriptions first earn returns and pay fees next month. Owner units stay fixed, with no personal withdrawals. [Monthly unit accounts: inflows.simulate](#).

Closing assets = opening assets + investment P/L - investor fee + subscriptions - redemptions

External-business cash = external fee receipts - fixed costs - acquisition costs - servicing costs - setup costs

The historical execution additionally marks portfolio prices, accrues financing and executes prior investment signals. It charges the fund fee on previous-event net fund equity and adjusts positions for cash flow before forming the next signal. Flow trades preserve the current mixture and use pre-cost exposure rather than resetting target weights. At borrowed exposure, flow-trading expenses are posted to debt; post-cost leverage can drift and change subsequent funding and signals. Exact post-cost leverage preservation is not assumed at commission discontinuities. [Fund units and flows: coupled_inflow_accounting.simulate_coupled](#).

3. No-External-Investor Reference

At the 5bp fee and without external subscriptions, hypothetical owner equity after ten years is USD 6.97 million under 0%, USD 12.47 million under 6%, USD 4.88 million under the early loss, and USD 5.81 million under the later loss. Under a fixed return path, external subscriptions do not change owner unit performance. This invariance is not imposed on the historical coupled portfolio, where fees and flow trades can alter later investment accounts.

[Client, fee and cost settings: inflow_acquisition.json](#) · [Acquisition accounts: inflow_scenarios.acquisition](#).

A commercial operation with zero clients still incurs the assumed business costs. That counterfactual does not describe the cost of a personal portfolio, which may avoid fund-business setup and sales infrastructure. Appendix B includes the historical no-flow/no-added-fund-fee control; ordinary investment trading and financing costs remain in it.

4. Historical Portfolio and Fund Results

4.1 The strategy is actually executed with flows

The historical results use the ± 5 -percentage-point portfolio and separate leverage management. They are not calculated by inserting the full-period backtest CAGR as a constant monthly return. The full investment backtest continues to August 2026, whereas the business experiment ends in October 2024. Horizon, the added fund fee and flow feedback make the headline return measures different.

An **overlay diagnostic** applies the no-flow portfolio's dated return path to cohorts without executing subscriptions, redemptions or fee financing through the holdings and debt account. Its unit return is invariant across plans at a fixed fee schedule. The coupled unit return need not be invariant.

Table 3. Steady acquisition: coupled execution and overlay · [Fund return comparison:](#)

[historical_inflow_interpretation.compare_accounts](#)

Rule	Coupled unit TWR / year	Overlay unit TWR / year	Coupled external MWR / year
Absolute / separate leverage	10.73%	11.08%	12.71%
Capped-relative comparison	10.61%	10.73%	12.59%

For the main absolute/separate strategy, coupled unit TWR differs from the overlay by -0.3459 pp. Direct flow-trading charges are USD 111,423.03. Those charges alone do not causally explain the difference; debt posting, funding and changed intervention paths also matter. Capped-relative remains a sensitivity comparison. [Fund return comparison: historical_inflow_interpretation.compare_accounts](#).

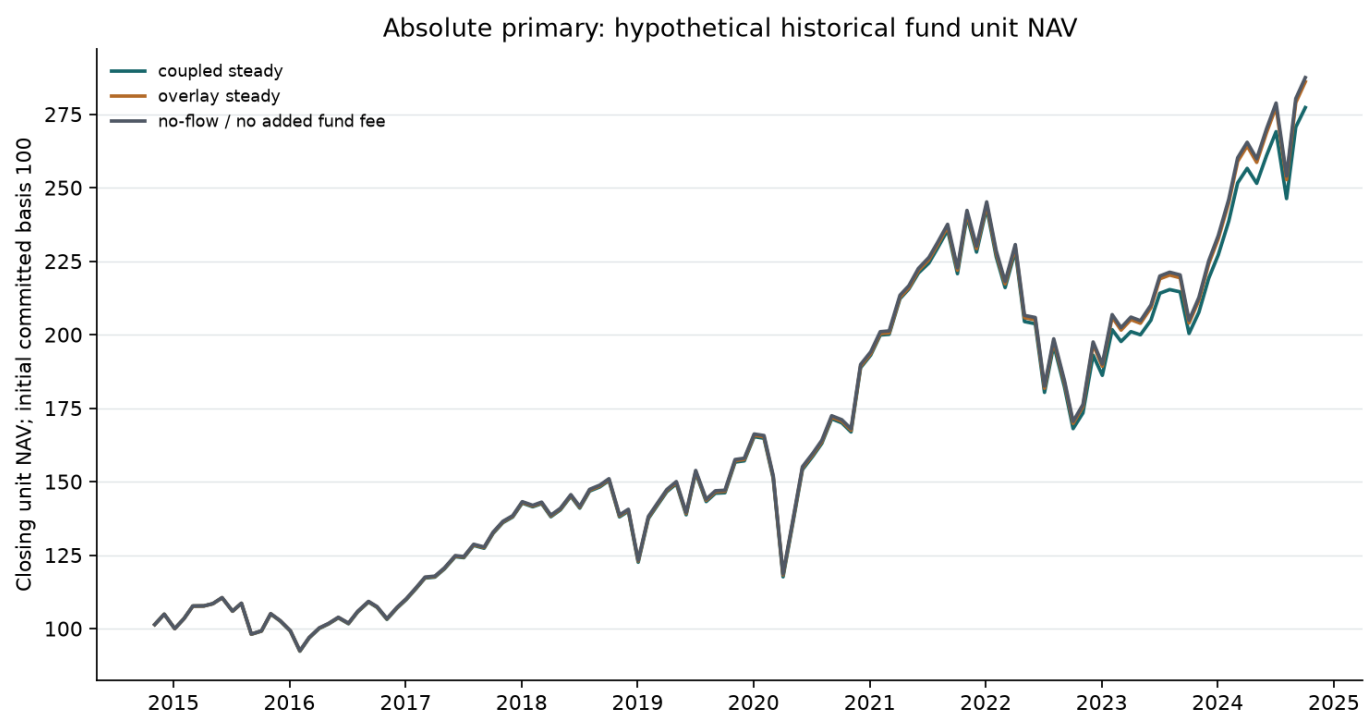


Figure 1. Historical monthly closing unit NAV for the main strategy: no-added-fund-fee investment control, steady coupled fund and overlay diagnostic. Opening investment costs remain included. The fee and client accounts are hypothetical, even though investment inputs are actual saved market observations. [Historical NAV plot: coordinated_inflow_report.historical_figures](#)

4.2 Ownership and cash already returned to investors

Table 4. Main strategy: historical acquisition outcomes, USD million · [Fund units and flows: coupled_inflow_accounting.simulate_coupled](#)

Acquisition plan	Owner invested equity, USD m	External closing equity, USD m	Total net fund assets, USD m	Cumulative redemptions, USD m
No external flows	19.60	0.00	19.60	0.00
Limited demand	19.57	4.29	23.86	1.70
Steady acquisition	19.41	29.43	48.84	10.00
Strong acquisition	19.38	58.84	78.22	20.00

The large asset totals include client-owned capital. Owner invested equity excludes outside manager cash and personal withdrawals. Redemptions are cash returned to investors and are excluded from closing external equity; they are not automatically destroyed wealth.

Fund-unit TWR measures the compounded unit performance. Aggregate external MWR also accounts for when clients subscribe and redeem. Subscriptions are negative investor cash flows; redemptions and terminal external equity are positive. Historical MWR uses actual dated transactions and elapsed days divided by 365.2425. The external-investor window begins at the first subscription and differs from the full fund window. No-investor MWR is unavailable, never zero. One root found by a finite scan does not prove global uniqueness or detection of every tangency. [Dated investor return: historical_inflow_diagnostics.dated_mwr](#).

4.3 Manager cash and working capital

Table 5. Main strategy, steady acquisition: historical manager accounts, USD · [Manager cash: historical_inflow_diagnostics.manager_accounts](#)

Receipt	External receipts, USD	Costs, USD	External-business cash, USD	External peak gap, USD	Conditional total-fee cash, USD
100.00%	47,180.72	455,115.31	-407,934.60	407,934.60	-351,319.87
50.00%	23,590.36	455,115.31	-431,524.95	431,524.95	-403,217.59

At full fee receipt, external receipts are USD 47,180.72 against USD 455,115.31 of business costs. Peak external-business funding need is USD 407,934.60. Including conditional owner-fee receipts changes manager treasury but does not create external profit; these fees are already deducted from owner invested equity. [Manager cash: historical_inflow_diagnostics.manager_accounts](#).

Across the retained historical grid, 48 of 48 manager labels have negative terminal external-business cash. The labels include repeated no-flow economics and controls, so this is not a count of independent observations or a probability of failure. The result challenges the assumed fee-and-cost combination, not every possible fund business. [Manager cash: historical_inflow_diagnostics.manager_accounts](#).

Historical conditional treasury applies the same receipt fraction to owner and external fees, assuming monthly availability. The hypothetical treasury sensitivity in Appendix A has its own full owner-fee receipt convention. Neither establishes actual bank-account funding without settlement and contract terms. Peak gaps are measured at month-end; intramonth needs are excluded.

5. Hypothetical Scenarios and Sensitivities

5.1 Raising assets does not establish profitable operations

Table 6. Acquisition under the hypothetical 6% path · [Acquisition accounts: inflow_scenarios.acquisition](#)

Case	Owner equity, year 10 (USD m)	External equity, year 10 (USD m)	Cumulative manager cash, ten years (USD)
No external flows	12.47	0.00	-298,743
Limited demand	12.47	3.33	-314,480
Steady acquisition	12.47	23.49	-410,903
Strong acquisition	12.47	46.98	-523,063
Sales stop / runoff	12.47	0.47	-305,940

Steady acquisition generates USD 27.50 million gross subscriptions and USD 42,233 cumulative external fees over ten years. Its peak business funding gap is USD 410,903; the strong schedule's gap is USD 523,063. Acquisition spending occurs on subscriptions, while a low annual fee is earned over the time capital remains invested. More clients can therefore increase working-capital needs under this cost structure. [Acquisition accounts: inflow_scenarios.acquisition](#).

5.2 The fee rate and the amount actually received

Table 7. Steady acquisition: fee sensitivity under 6% · [Fee scenarios: inflow_results.sensitivity_tables](#)

Annual fee	Owner equity, year 10 (USD m)	Manager result, year 10 (USD)	Ten-year manager cash (USD)
0.05%	12.47	-48,474	-410,903
0.10%	12.41	-37,827	-368,749
0.25%	12.23	-6,078	-242,884
0.50%	11.93	46,205	-35,077

Higher fees reduce owner invested wealth. They also improve manager receipts if client counts, redemptions and receipt terms stay fixed; that fixed-demand assumption is not evidence that clients accept the higher charge. A positive final-year manager result does not imply recovery of startup and earlier losses.

At 50bp, the USD 35,077 closing deficit follows a USD 136,920 peak deficit in month 70. Working capital must cover the lowest balance on the path rather than only the final year. The conditional owner-transfer comparison is shown in Appendix A. [Sensitivity settings: inflow_controls.json](#) · [Loss timing: inflow_controls.loss_controls](#).

At the baseline fee and servicing assumptions, full receipt leaves 3bp of external assets for fixed and acquisition costs; 50% receipt leaves 0.5bp. The full sensitivity materials also vary tickets, receipt fractions and fixed/acquisition/servicing budgets. These adjustable inputs are not provider quotes or approved fee terms. [Net fee arithmetic: inflow_report.prose_claims](#) · [Client, fee and cost settings: inflow_acquisition.json](#).

5.3 Equal gross subscriptions at different times

A controlled experiment distributes USD 27.5 million of gross subscriptions across five schedules: constant monthly, linearly increasing monthly, even tranches, early tranches and late tranches. Subscriptions begin no earlier than month 7. This matches gross fundraising rather than net flows or closing assets. [Subscription schedules: inflow_timing.json](#) · [Client, fee and cost settings: inflow_acquisition.json](#).

Table 8. Selected timing cases under 6% · [Subscription timing: inflow_scenarios.timing](#)

Timing case	Total fund assets, year 10 (USD m)	Cumulative external redemptions (USD m)	Ten-year external fees (USD)
Constant monthly	34.64	11.83	56,139
Early steps	31.84	18.05	85,635
Late steps	37.04	6.51	30,867

Earlier cohorts pay fees for longer and face more redemption attrition. Later subscriptions can leave more assets in the fund at the endpoint while providing less earlier fee income. Cash already redeemed remains an investor outcome, so higher closing AUM does not by itself establish higher investor wealth. Owner invested equity is USD 12.47 million in all these fixed-return cases. [Acquisition accounts: inflow_scenarios.acquisition](#).

For the later-loss path, 6% in years 1-3, -30% in year 4, then 0%, unit TWR is identical across timing schedules while external MWR differs. Appendix B shows representative return measures. Late subscriptions avoid much of the stipulated loss, but clients cannot infer future loss dates from this retrospective illustration. [Client, fee and cost settings: inflow_acquisition.json](#).

Moving the same net annual loss into different individual months changes the subscription cohorts exposed to it. The early-tranche terminal-asset range is USD 836,556.50 despite unchanged annual net unit performance. Appendix B retains selected ranges. Smoothing losses across a year therefore suppresses a material timing sensitivity; these deterministic placements are not probabilities or worst-case limits. [Loss timing: inflow_controls.loss_controls](#).

6. Discussion and Conclusion

The study separates investment outcomes from fundraising and business economics. The historical branch executes the actual strategy with assumed investor flows and an added fund fee; the hypothetical branches isolate adjustable return, demand and timing assumptions. Their horizons and accounting should remain distinct.

External subscriptions can enlarge fund assets without becoming owner wealth. At the assumed low fee and the stated costs, the retained historical manager accounts require additional working capital. Strong fundraising alone does not resolve that shortfall. Fee retention, acquisition spending and servicing costs must be discussed together.

The next substantive inputs are evidenced client counts, tickets, redemption behaviour, fee-sharing contracts, provider invoices and the source of manager working capital. These would support a recalculation of an explicit business design. Historical investment performance cannot supply missing commercial evidence or executable credit terms. Personal withdrawals, tax, inflation and retirement sufficiency require a separate owner-wealth analysis.

The companion portfolio backtest presents investment performance, the MSCI World benchmark and brief Dimensional and Amundi comparisons. It contains no investor subscriptions or outside-fund manager revenue.

Sources and Data

Fund identities: Core IE00B4L5Y983; Momentum IE00BP3QZ825; Quality IE00BP3QZ601; Value IE00BP3QZB59.

Market data and financing series

NAV observations and reference rates enter the calculations. Their sample dates are specified in the analysis. Provider links identify the funds and series; the supplied local files are the calculation inputs.

Source	Publisher	Use in this report
Core MSCI World NAV	BlackRock / iShares	Core accumulating USD NAV; investment returns
Momentum NAV	BlackRock / iShares	Momentum accumulating USD NAV; investment returns
Quality NAV	BlackRock / iShares	Quality accumulating USD NAV; investment returns
Value NAV	BlackRock / iShares	Value accumulating USD NAV; investment returns
Indicative reference rate	Federal Reserve Bank of New York	Early indicative financing reference series
Official SOFR	Federal Reserve Bank of New York	Later official financing reference series

Cost, product and methodology references

These sources support the stated assumptions, product descriptions and checks. They are distinguished from time-series inputs.

Source	Publisher	Use in this report
Commission tariff	Swissquote	Current CHF commission schedule for the investment cost model
Investor-side stamp duty	Swiss Federal Tax Administration	Basis for the modelled 0.15% investor-side duty
Commission FX conversion	ECB / Official Journal	31 August 2026 reference cross-rate; CHF commission conversion only
Fund fee and expense terminology	Investor.gov / U.S. SEC	Background on fund fees and expenses; business assumptions are author-defined

Original files and download records are retained with the reproducibility materials. Fund-business assumptions about client acquisition, redemptions, fees and budgets are author-defined scenarios rather than observed fundraising data.

Appendix A. Hypothetical Scenario Overview

Table A1. Closing total fund assets under all twenty acquisition combinations, USD million · [Acquisition accounts: inflow scenarios.acquisition](#)

Flow case	Flat	6%	Early loss	Later loss
No external flows	6.97	12.47	4.88	5.81
Limited demand	9.66	15.80	7.57	8.26
Steady acquisition	26.67	35.96	24.47	24.35
Strong acquisition	46.37	59.45	44.07	42.89
Sales stop / runoff	7.24	12.94	5.08	6.00

Columns distinguish flat, fixed growth, early loss and later loss. Acquisition loss cases include halved subscriptions and higher attrition. These totals include owner capital and external equity; they are not consolidated investor wealth or probabilities.

Table A2. External-business funding and conditional owner-fee receipts, USD · [Funding gaps: inflow treasury.treasury_controls](#)

Annual fee	Closing cash, external fees only (USD)	Peak deficit, external fees only (USD)	Closing cash, including owner fees (USD)	Peak deficit, including owner fees (USD)
0.05%	-410,903	410,903	-363,643	363,643
0.10%	-368,749	368,749	-274,485	274,485
0.25%	-242,884	242,884	-9,124	64,201
0.50%	-35,077	136,920	426,190	24,167

The hypothetical illustration assumes 100% internal owner-fee receipt at monthly accrual, no redistribution to owners, tax or settlement delay. These internal transfers are already deducted from invested owner equity and must not be counted twice as consolidated wealth.

Appendix B. Return and Timing Controls

Table B1. Aggregate external investor returns under the hypothetical later loss · [Investor returns: inflow_returns.external_investor_returns](#)

Timing case	Annualised unit TWR	Annualised aggregate external MWR
Constant monthly	-1.85%	-2.06%
Early steps	-1.85%	-2.99%
Late steps	-1.85%	-0.05%

Hypothetical MWR solves equal-month cash-flow IRR and annualises over twelve months. It is not dated transaction XIRR or an individual client's return. Timing schedules and 10% annual unit attrition stay fixed in these loss cases, unlike the acquisition demand stress. [Cash-flow IRR: inflow_returns.money_weighted_return](#).

Table B2. Within-year loss placement with matched annual net unit performance · [Loss timing: inflow_controls.loss_controls](#)

Loss-placement control	Minimum ending total assets (USD)	Maximum ending total assets (USD)	Range (USD)
Steady acquisition	24,288,550.90	24,410,973.79	122,422.89
Constant monthly	22,062,274.73	22,462,487.44	400,212.70
Early steps	16,008,421.70	16,844,978.19	836,556.50

The loss is concentrated into each possible month while preserving the same net annual unit multiplier. With monthly fee f and smooth return r , the matched multiplier is $G = (1 + r - f)^{12}$; the single gross shock is $G / (1 - f)^{11} - 1 + f$. Subscriptions and each layer's redemption convention remain fixed within the control. Ranges use unrounded endpoints and do not bound actual markets. [Loss timing: inflow_controls.loss_controls](#).

Table B3. Historical no-flow/no-added-fund-fee controls · [Historical controls: historical_inflow_replay.calculate_replays](#)

Rule	Owner ending equity, USD m	Added fund fee, USD	Investment trading cost, USD	Financing, USD
Absolute / separate leverage	20.12	0.00	26,158.23	1,417,942.70
Capped-relative comparison	19.49	0.00	28,032.34	1,234,045.16

The controls retain ordinary investment entry, trading and financing costs. They reconcile to the investment paths over the shorter business horizon. No external cohort is invested, so external MWR is unavailable.

Reproducibility Materials

Blue highlighted links name the source function or configuration and the quantity it calculates or specifies. Click a function link to open the corresponding code line on GitHub. Code links identify a fixed source snapshot; [the report-to-calculation map](#) provides the selected rows, fields and generated evidence paths. Original provider observations remain local.

The prepared code, explicit business configurations and separately supplied original investment data reconstruct the accounts. Complete dated historical returns, root-search statuses, monthly cohorts, timing schedules, fee/ticket/receipt and cost grids, additional figures and reconciliation records remain accompanying calculation material. They preserve adjustable assumptions and omitted evidence without lengthening the main argument. The reader-selection record identifies every displayed result's source.

AI Assistance and Responsibility

This portfolio-backtest and capital-inflow research package was developed throughout with extensive assistance from generative artificial intelligence. AI was used across the analytical and written workflow, including methodology development, programming, data processing, backtesting and scenario analysis, statistical evaluation, interpretation, report drafting and editing, tables, figures, and documentation. The original market data and external documents come from their respective providers and are referenced separately. The accompanying reproducibility materials document the calculations, assumptions, outputs, and checks. Responsibility for the submitted content remains with the human author(s).

Scope of assistance

OpenAI Codex was used for substantive code generation and revision, methodological discussion, analysis and report drafting. AI-generated or substantially revised wording appears throughout the abstract, main text, source commentary and appendices. Tables and figures were produced by AI-assisted code using the stated data and scenario assumptions. Provider data and external documents retain their original attribution.

List of aids (Hilfsmittelverzeichnis)

Tool	Purpose and affected material
OpenAI Codex (OpenAI, n.d.)	Method development, code generation/revision, data processing, backtests and scenarios, statistics, interpretation, all report prose, tables, figures and documentation
Python, NumPy, pandas and openpyxl	Numerical implementation, data alignment/accounting, scenario processing, original spreadsheet readers and report evidence
Matplotlib	Programmatic charts from calculated investment and fund paths
pytest	Automated calculation and report-contract checks
pypdf, pdfplumber, ReportLab, Pillow and Poppler	Source-PDF extraction/checks, report PDF generation and rendered-page verification

Author contribution and verification

Alex specified and approved key design choices, supplied project material, determined which comparisons to present and reviewed successive drafts. Codex helped translate these instructions into data readers, portfolio and fund simulations, analytical controls and reporting routines. It also contributed to the development of methods and interpretation of results.

Verification covered accounting identities, comparisons with retained reference outcomes, separate implementations, report evidence bindings and numerical, text and layout checks. These checks assess implementation and reporting under shared inputs and conventions; they do not constitute independent human expert review. The accompanying evidence record identifies the checks performed and their limits.

Documentation and archive

The accompanying AI-use disclosure record summarises the assistance, author contribution and verification limits. Selected exact prompt evidence and detailed internal working records are retained locally for applicable assessment/archive or review requirements; they are not distributed with the public reports.

A complete chronological prompt/output archive has not been established. Codex is identified at application level; no historical model version is asserted. Factual claims refer to the separately cited original evidence. The author remains responsible for the content and any assessment-specific documentation obligations.

AI application reference

OpenAI. (n.d.). *Codex* [Generative AI coding assistant]. Application reference.

This declaration concerns Alex's research contribution. It does not attribute unverified AI use to other group members or future video narration.